In [2]:
```
# Import libraries
import matplotlib.pyplot as plt
from matplotlib.patches import Patch
import pandas as pd
import datetime
import sqlite3
import numpy as np
from scipy import stats
from itertools import combinations
from statsmodels.stats.multitest import multipletests

pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
```

Import the data

> https://www.transtats.bts.gov/OT_Delay/OT_DelayCause1.asp

Navigate to "Database Tables" > Reporting Carrier On-Time Performance (1987-present)

Select all fields from Year to DivDistance (inclusive) and download the dataset for August 2025 - July 2026

Alternatively, unzip data.zip and place all the csv files in /data

In [3]:
```
files = [ "08_2025", "09_2025", "10_2025", "11_2025", "12_2025", "01_2026", "02_202
flights_df = pd.DataFrame()

for name in files:
    print("Loading", name)
    loaded_df = pd.read_csv(f'./data/{name}.csv', dtype={48: str})
    flights_df = pd.concat([flights_df, loaded_df], verify_integrity=False)

flights_df.to_csv("./data/flights.csv")
```

```
Loading 08_2025
Loading 09_2025
Loading 10_2025
Loading 11_2025
Loading 12_2025
Loading 01_2026
Loading 02_2026
Loading 03_2026
Loading 04_2026
Loading 05_2026
Loading 06_2026
Loading 07_2026
```

Cleaning the Data

First we need to load our data. Then, we can take a look at what values are NaN/missing.

In [4]:
```
flights_df = pd.read_csv("./data/flights.csv", dtype={49: str}, index_col=0)

display(flights_df.head(5))
flights_df.isna().sum()
```

	YEAR	QUARTER	MONTH	DAY_OF_MONTH	DAY_OF_WEEK	FL_DATE	OP_UNIQUE_CARRI
0	2025	3	8	1	5	8/1/2025 12:00:00 AM	A
1	2025	3	8	1	5	8/1/2025 12:00:00 AM	A
2	2025	3	8	1	5	8/1/2025 12:00:00 AM	A
3	2025	3	8	1	5	8/1/2025 12:00:00 AM	A
4	2025	3	8	1	5	8/1/2025 12:00:00 AM	A

Out[4]:

YEAR	0
QUARTER	0
MONTH	0
DAY_OF_MONTH	0
DAY_OF_WEEK	0
FL_DATE	0
OP_UNIQUE_CARRIER	0
OP_CARRIER_AIRLINE_ID	0
OP_CARRIER	0
TAIL_NUM	17774
OP_CARRIER_FL_NUM	0
ORIGIN_AIRPORT_ID	0
ORIGIN_AIRPORT_SEQ_ID	0
ORIGIN_CITY_MARKET_ID	0
ORIGIN	0
ORIGIN_CITY_NAME	0
ORIGIN_STATE_ABR	0
ORIGIN_STATE_FIPS	0
ORIGIN_STATE_NM	0
ORIGIN_WAC	0
DEST_AIRPORT_ID	0
DEST_AIRPORT_SEQ_ID	0
DEST_CITY_MARKET_ID	0
DEST	0
DEST_CITY_NAME	0
DEST_STATE_ABR	0
DEST_STATE_FIPS	0
DEST_STATE_NM	0
DEST_WAC	0
CRS_DEP_TIME	0
DEP_TIME	123266
DEP_DELAY	123827
DEP_DELAY_NEW	123827
DEP_DEL15	123827
DEP_DELAY_GROUP	123827
DEP_TIME_BLK	0
TAXI_OUT	127718
WHEELS_OFF	127718
WHEELS_ON	130741
TAXI_IN	130741
CRS_ARR_TIME	0
ARR_TIME	130741
ARR_DELAY	148348
ARR_DELAY_NEW	148348
ARR_DEL15	148348
ARR_DELAY_GROUP	148348
ARR_TIME_BLK	0
CANCELLED	0
CANCELLATION_CODE	6915477
DIVERTED	0
CRS_ELAPSED_TIME	2
ACTUAL_ELAPSED_TIME	148348
AIR_TIME	148348
FLIGHTS	0
DISTANCE	0
DISTANCE_GROUP	0

```
CARRIER_DELAY              5501363
WEATHER_DELAY              5501363
NAS_DELAY                  5501363
SECURITY_DELAY             5501363
LATE_AIRCRAFT_DELAY        5501363
FIRST_DEP_TIME             6990179
TOTAL_ADD_GTIME            6990194
LONGEST_ADD_GTIME          6990194
DIV_AIRPORT_LANDINGS             2
DIV_REACHED_DEST           7023892
DIV_ACTUAL_ELAPSED_TIME    7026251
DIV_ARR_DELAY              7026251
DIV_DISTANCE               7023892
dtype: int64
```

We are examining flights based on their departure and arrival time, so if neither are available we can safely discard them since no data was recorded. There are only a small number of these (< 1%)

We can also fix the delay type fields since they are filled with NaN instead of 0 for no delay.

In [5]:
```
flights_df.dropna(subset=['DEP_TIME', 'ARR_TIME'], inplace=True) # Drop flights wit
flights_df['CARRIER_DELAY'] = flights_df['CARRIER_DELAY'].fillna(0) # NaN is no del
flights_df['WEATHER_DELAY'] = flights_df['WEATHER_DELAY'].fillna(0) # NaN is no del
flights_df['NAS_DELAY'] = flights_df['NAS_DELAY'].fillna(0) # NaN is no delay
flights_df['SECURITY_DELAY'] = flights_df['SECURITY_DELAY'].fillna(0) # NaN is no d
flights_df['LATE_AIRCRAFT_DELAY'] = flights_df['LATE_AIRCRAFT_DELAY'].fillna(0) # N

print(flights_df.isna().sum())
display(flights_df[flights_df['DIV_AIRPORT_LANDINGS'].isna()])
print(flights_df['DIV_AIRPORT_LANDINGS'].value_counts(dropna=False))
```

YEAR	0
QUARTER	0
MONTH	0
DAY_OF_MONTH	0
DAY_OF_WEEK	0
FL_DATE	0
OP_UNIQUE_CARRIER	0
OP_CARRIER_AIRLINE_ID	0
OP_CARRIER	0
TAIL_NUM	0
OP_CARRIER_FL_NUM	0
ORIGIN_AIRPORT_ID	0
ORIGIN_AIRPORT_SEQ_ID	0
ORIGIN_CITY_MARKET_ID	0
ORIGIN	0
ORIGIN_CITY_NAME	0
ORIGIN_STATE_ABR	0
ORIGIN_STATE_FIPS	0
ORIGIN_STATE_NM	0
ORIGIN_WAC	0
DEST_AIRPORT_ID	0
DEST_AIRPORT_SEQ_ID	0
DEST_CITY_MARKET_ID	0
DEST	0
DEST_CITY_NAME	0
DEST_STATE_ABR	0
DEST_STATE_FIPS	0
DEST_STATE_NM	0
DEST_WAC	0
CRS_DEP_TIME	0
DEP_TIME	0
DEP_DELAY	0
DEP_DELAY_NEW	0
DEP_DEL15	0
DEP_DELAY_GROUP	0
DEP_TIME_BLK	0
TAXI_OUT	0
WHEELS_OFF	0
WHEELS_ON	0
TAXI_IN	0
CRS_ARR_TIME	0
ARR_TIME	0
ARR_DELAY	17607
ARR_DELAY_NEW	17607
ARR_DEL15	17607
ARR_DELAY_GROUP	17607
ARR_TIME_BLK	0
CANCELLED	0
CANCELLATION_CODE	6913117
DIVERTED	0
CRS_ELAPSED_TIME	2
ACTUAL_ELAPSED_TIME	17607
AIR_TIME	17607
FLIGHTS	0
DISTANCE	0
DISTANCE_GROUP	0

```
CARRIER_DELAY                   0
WEATHER_DELAY                   0
NAS_DELAY                       0
SECURITY_DELAY                  0
LATE_AIRCRAFT_DELAY             0
FIRST_DEP_TIME            6863990
TOTAL_ADD_GTIME           6864002
LONGEST_ADD_GTIME         6864002
DIV_AIRPORT_LANDINGS            1
DIV_REACHED_DEST          6895510
DIV_ACTUAL_ELAPSED_TIME   6895510
DIV_ARR_DELAY             6895510
DIV_DISTANCE              6895510
dtype: int64
```

	YEAR	QUARTER	MONTH	DAY_OF_MONTH	DAY_OF_WEEK	FL_DATE	OP_UNIQUE_C/
27791	2026	3	7	2	4	7/2/2026 12:00:00 AM	

```
DIV_AIRPORT_LANDINGS
0.0    6895391
1.0      17501
9.0        118
2.0        104
3.0          2
NaN          1
Name: count, dtype: int64
```

There is still one flight that has a NaN number of airport landings. It looks like this flight has no diversion data anyways, so it did not divert. The standard value for this is 0, not NaN, so we can correct it.

In [6]:
```
flights_df['DIV_AIRPORT_LANDINGS'] = flights_df['DIV_AIRPORT_LANDINGS'].fillna(0)
print(flights_df['DIV_AIRPORT_LANDINGS'].value_counts(dropna=False))
```

```
DIV_AIRPORT_LANDINGS
0.0    6895392
1.0      17501
9.0        118
2.0        104
3.0          2
Name: count, dtype: int64
```

Let's parse our data now! First, we need to look at what types we're working with.

In [7]:
```
print(flights_df.dtypes)
display(flights_df.sample(10))
```

```
YEAR                      int64
QUARTER                   int64
MONTH                     int64
DAY_OF_MONTH              int64
DAY_OF_WEEK               int64
FL_DATE                  object
OP_UNIQUE_CARRIER        object
OP_CARRIER_AIRLINE_ID     int64
OP_CARRIER               object
TAIL_NUM                 object
OP_CARRIER_FL_NUM         int64
ORIGIN_AIRPORT_ID         int64
ORIGIN_AIRPORT_SEQ_ID     int64
ORIGIN_CITY_MARKET_ID     int64
ORIGIN                   object
ORIGIN_CITY_NAME         object
ORIGIN_STATE_ABR         object
ORIGIN_STATE_FIPS         int64
ORIGIN_STATE_NM          object
ORIGIN_WAC                int64
DEST_AIRPORT_ID           int64
DEST_AIRPORT_SEQ_ID       int64
DEST_CITY_MARKET_ID       int64
DEST                     object
DEST_CITY_NAME           object
DEST_STATE_ABR           object
DEST_STATE_FIPS           int64
DEST_STATE_NM            object
DEST_WAC                  int64
CRS_DEP_TIME              int64
DEP_TIME                float64
DEP_DELAY               float64
DEP_DELAY_NEW           float64
DEP_DEL15               float64
DEP_DELAY_GROUP         float64
DEP_TIME_BLK             object
TAXI_OUT                float64
WHEELS_OFF              float64
WHEELS_ON               float64
TAXI_IN                 float64
CRS_ARR_TIME              int64
ARR_TIME                float64
ARR_DELAY               float64
ARR_DELAY_NEW           float64
ARR_DEL15               float64
ARR_DELAY_GROUP         float64
ARR_TIME_BLK             object
CANCELLED               float64
CANCELLATION_CODE        object
DIVERTED                float64
CRS_ELAPSED_TIME        float64
ACTUAL_ELAPSED_TIME     float64
AIR_TIME                float64
FLIGHTS                 float64
DISTANCE                float64
DISTANCE_GROUP            int64
```

```
CARRIER_DELAY              float64
WEATHER_DELAY              float64
NAS_DELAY                  float64
SECURITY_DELAY             float64
LATE_AIRCRAFT_DELAY        float64
FIRST_DEP_TIME             float64
TOTAL_ADD_GTIME            float64
LONGEST_ADD_GTIME          float64
DIV_AIRPORT_LANDINGS       float64
DIV_REACHED_DEST           float64
DIV_ACTUAL_ELAPSED_TIME    float64
DIV_ARR_DELAY              float64
DIV_DISTANCE               float64
dtype: object
```

	YEAR	QUARTER	MONTH	DAY_OF_MONTH	DAY_OF_WEEK	FL_DATE	OP_UNIQU
521022	2026	3	7	26	7	7/26/2026 12:00:00 AM	
496529	2026	2	6	25	4	6/25/2026 12:00:00 AM	
348135	2025	4	12	19	5	12/19/2025 12:00:00 AM	
50739	2026	1	2	3	2	2/3/2026 12:00:00 AM	
214757	2025	4	12	12	5	12/12/2025 12:00:00 AM	
246401	2025	4	10	13	1	10/13/2025 12:00:00 AM	
401540	2026	3	7	20	1	7/20/2026 12:00:00 AM	
303710	2025	3	9	17	3	9/17/2025 12:00:00 AM	
117976	2025	4	11	7	5	11/7/2025 12:00:00 AM	
223845	2026	2	5	12	2	5/12/2026 12:00:00 AM	

Our FL_DATE is a string, not a date object. It also does not include the departure time. Let's convert it using the individual components after we verify their ranges make sense, and add it to a column named "SCHEDULED_DATE". We should use CRS_DEP_TIME (formatted hhmm) since we want to know the scheduled time before any delays.

Unfortunately, this data is also messy and does not follow the format exactly. Instead, we will need to format the values (1 to 2359) by padding them with leading 0s to match the format. For example: 1 should become 0001 (or 00:01)

In [8]:
```
def create_date(row):
    return datetime.datetime(row['YEAR'], row['MONTH'], row['DAY_OF_MONTH'], hour=i

# Look at current distribution
print("Departure Time:", flights_df['CRS_DEP_TIME'].min(), "-", flights_df['CRS_DEP
print("Year:", flights_df['YEAR'].min(), "-", flights_df['YEAR'].max())
print("Month:", flights_df['MONTH'].min(), "-", flights_df['MONTH'].max())
print("Day:", flights_df['DAY_OF_MONTH'].min(), "-", flights_df['DAY_OF_MONTH'].max

# Clean up the departure time
flights_df['CRS_DEP_TIME'] = flights_df['CRS_DEP_TIME'].astype(str).apply(lambda x:
print(flights_df['CRS_DEP_TIME'].sample(10))

# Convert to date
flights_df.drop(columns=['FL_DATE'], inplace=True)
flights_df['DATE'] = flights_df.apply(create_date, axis=1)
display(flights_df[['DATE', 'CRS_DEP_TIME']].sample(10))
```

```
Departure Time: 1 - 2359
Year: 2025 - 2026
Month: 1 - 12
Day: 1 - 31
227586    1812
37482     1400
262538    1839
307987    1630
526491    1720
509898    1100
232096    0540
392992    1110
354851    1259
30688     0715
Name: CRS_DEP_TIME, dtype: object
```

	DATE	CRS_DEP_TIME
35374	2025-09-02 18:50:00	1850
325776	2026-07-17 14:24:00	1424
216955	2025-08-11 07:45:00	0745
548377	2025-08-29 14:11:00	1411
390365	2026-06-20 18:16:00	1816
25269	2026-06-02 15:10:00	1510
332116	2025-08-17 11:17:00	1117
220706	2026-01-13 17:59:00	1759
448249	2026-02-25 15:33:00	1533
215275	2026-07-11 18:10:00	1810

The rest of the data is parsed correctly, so we can continue with exporting this to a database!

Exporting to a Database

We will now export all of our data to an sqlite file using `to_sql` to the "Flights" table

In [9]:
```
conn = sqlite3.connect("database.sqlite")

flights_df.to_sql(name='Flights', con=conn, if_exists='replace')
```

Out[9]: 6913117

Let's verify that our table is setup correctly:

In [10]:
```
crsr = conn.cursor()
query = "PRAGMA table_info(Flights)"
print(crsr.execute(query).fetchall())

query = "SELECT COUNT(*) FROM Flights"
print("Database Rows:", crsr.execute(query).fetchall()[0][0])
print("DataFrame Rows:", flights_df.shape[0])
```

```
[(0, 'index', 'INTEGER', 0, None, 0), (1, 'YEAR', 'INTEGER', 0, None, 0), (2, 'QUART
ER', 'INTEGER', 0, None, 0), (3, 'MONTH', 'INTEGER', 0, None, 0), (4, 'DAY_OF_MONT
H', 'INTEGER', 0, None, 0), (5, 'DAY_OF_WEEK', 'INTEGER', 0, None, 0), (6, 'OP_UNIQU
E_CARRIER', 'TEXT', 0, None, 0), (7, 'OP_CARRIER_AIRLINE_ID', 'INTEGER', 0, None,
0), (8, 'OP_CARRIER', 'TEXT', 0, None, 0), (9, 'TAIL_NUM', 'TEXT', 0, None, 0), (10,
'OP_CARRIER_FL_NUM', 'INTEGER', 0, None, 0), (11, 'ORIGIN_AIRPORT_ID', 'INTEGER', 0,
None, 0), (12, 'ORIGIN_AIRPORT_SEQ_ID', 'INTEGER', 0, None, 0), (13, 'ORIGIN_CITY_MA
RKET_ID', 'INTEGER', 0, None, 0), (14, 'ORIGIN', 'TEXT', 0, None, 0), (15, 'ORIGIN_C
ITY_NAME', 'TEXT', 0, None, 0), (16, 'ORIGIN_STATE_ABR', 'TEXT', 0, None, 0), (17,
'ORIGIN_STATE_FIPS', 'INTEGER', 0, None, 0), (18, 'ORIGIN_STATE_NM', 'TEXT', 0, Non
e, 0), (19, 'ORIGIN_WAC', 'INTEGER', 0, None, 0), (20, 'DEST_AIRPORT_ID', 'INTEGER',
0, None, 0), (21, 'DEST_AIRPORT_SEQ_ID', 'INTEGER', 0, None, 0), (22, 'DEST_CITY_MAR
KET_ID', 'INTEGER', 0, None, 0), (23, 'DEST', 'TEXT', 0, None, 0), (24, 'DEST_CITY_N
AME', 'TEXT', 0, None, 0), (25, 'DEST_STATE_ABR', 'TEXT', 0, None, 0), (26, 'DEST_ST
ATE_FIPS', 'INTEGER', 0, None, 0), (27, 'DEST_STATE_NM', 'TEXT', 0, None, 0), (28,
'DEST_WAC', 'INTEGER', 0, None, 0), (29, 'CRS_DEP_TIME', 'TEXT', 0, None, 0), (30,
'DEP_TIME', 'REAL', 0, None, 0), (31, 'DEP_DELAY', 'REAL', 0, None, 0), (32, 'DEP_DE
LAY_NEW', 'REAL', 0, None, 0), (33, 'DEP_DEL15', 'REAL', 0, None, 0), (34, 'DEP_DELA
Y_GROUP', 'REAL', 0, None, 0), (35, 'DEP_TIME_BLK', 'TEXT', 0, None, 0), (36, 'TAXI_
OUT', 'REAL', 0, None, 0), (37, 'WHEELS_OFF', 'REAL', 0, None, 0), (38, 'WHEELS_ON',
'REAL', 0, None, 0), (39, 'TAXI_IN', 'REAL', 0, None, 0), (40, 'CRS_ARR_TIME', 'INTE
GER', 0, None, 0), (41, 'ARR_TIME', 'REAL', 0, None, 0), (42, 'ARR_DELAY', 'REAL',
0, None, 0), (43, 'ARR_DELAY_NEW', 'REAL', 0, None, 0), (44, 'ARR_DEL15', 'REAL', 0,
None, 0), (45, 'ARR_DELAY_GROUP', 'REAL', 0, None, 0), (46, 'ARR_TIME_BLK', 'TEXT',
0, None, 0), (47, 'CANCELLED', 'REAL', 0, None, 0), (48, 'CANCELLATION_CODE', 'TEX
T', 0, None, 0), (49, 'DIVERTED', 'REAL', 0, None, 0), (50, 'CRS_ELAPSED_TIME', 'REA
L', 0, None, 0), (51, 'ACTUAL_ELAPSED_TIME', 'REAL', 0, None, 0), (52, 'AIR_TIME',
'REAL', 0, None, 0), (53, 'FLIGHTS', 'REAL', 0, None, 0), (54, 'DISTANCE', 'REAL',
0, None, 0), (55, 'DISTANCE_GROUP', 'INTEGER', 0, None, 0), (56, 'CARRIER_DELAY', 'R
EAL', 0, None, 0), (57, 'WEATHER_DELAY', 'REAL', 0, None, 0), (58, 'NAS_DELAY', 'REA
L', 0, None, 0), (59, 'SECURITY_DELAY', 'REAL', 0, None, 0), (60, 'LATE_AIRCRAFT_DEL
AY', 'REAL', 0, None, 0), (61, 'FIRST_DEP_TIME', 'REAL', 0, None, 0), (62, 'TOTAL_AD
D_GTIME', 'REAL', 0, None, 0), (63, 'LONGEST_ADD_GTIME', 'REAL', 0, None, 0), (64,
'DIV_AIRPORT_LANDINGS', 'REAL', 0, None, 0), (65, 'DIV_REACHED_DEST', 'REAL', 0, Non
e, 0), (66, 'DIV_ACTUAL_ELAPSED_TIME', 'REAL', 0, None, 0), (67, 'DIV_ARR_DELAY', 'R
EAL', 0, None, 0), (68, 'DIV_DISTANCE', 'REAL', 0, None, 0), (69, 'DATE', 'TIMESTAM
P', 0, None, 0)]
Database Rows: 6913117
DataFrame Rows: 6913117
```

Now that our data is in our database, we can cleanup the dataframe from memory and continue!

In [11]:
```
flights_df = None
```

Data Exploration & Summary Statistics

Exploring Arrival Delays

We will explore arrival delays (arr_delay), the delay people really care about when their flight is delayed.

In [12]:
```
conn = sqlite3.connect("database.sqlite")
crsr = conn.cursor()

query = "SELECT COUNT(arr_delay) FROM Flights"
crsr.execute(query).fetchall()
```

Out[12]:
```
[(6895510,)]
```

This is a lot of entries! Let's take a look at what's inside.

In [13]:
```
flights_df = pd.read_sql_query("SELECT arr_delay, diverted, op_carrier FROM Flights

print(flights_df.describe())
flights_df[flights_df['ARR_DELAY'].isna()].head(10)
```

```
         ARR_DELAY      DIVERTED
count  6.895510e+06  6.913117e+06
mean   8.111811e+00  2.546897e-03
std    6.065110e+01  5.040249e-02
min   -1.090000e+02  0.000000e+00
25%   -1.600000e+01  0.000000e+00
50%   -6.000000e+00  0.000000e+00
75%    1.100000e+01  0.000000e+00
max    4.336000e+03  1.000000e+00
```

Out[13]:

	ARR_DELAY	DIVERTED	OP_CARRIER
292	NaN	1.0	AA
416	NaN	1.0	AA
604	NaN	1.0	AA
721	NaN	1.0	AA
962	NaN	1.0	AA
980	NaN	1.0	AA
1043	NaN	1.0	AA
1077	NaN	1.0	AA
1168	NaN	1.0	AA
1548	NaN	1.0	AA

We have a problem, some of the arrival delays are NaN! In this dataset, this looks related to the flight being diverted and not arriving at its destination. Let's confirm.

In [14]:
```
flights_df[(flights_df['ARR_DELAY'].isna()) & (flights_df['DIVERTED'] != 1)].head(1
```

Out[14]:

	ARR_DELAY	DIVERTED	OP_CARRIER

This theory is correct! For our analysis of the delay, let's only look at flights that actually arrived.

In [15]:
```
flights_df = pd.read_sql_query("SELECT arr_delay, diverted, op_carrier FROM Flights
flights_df.describe()
```

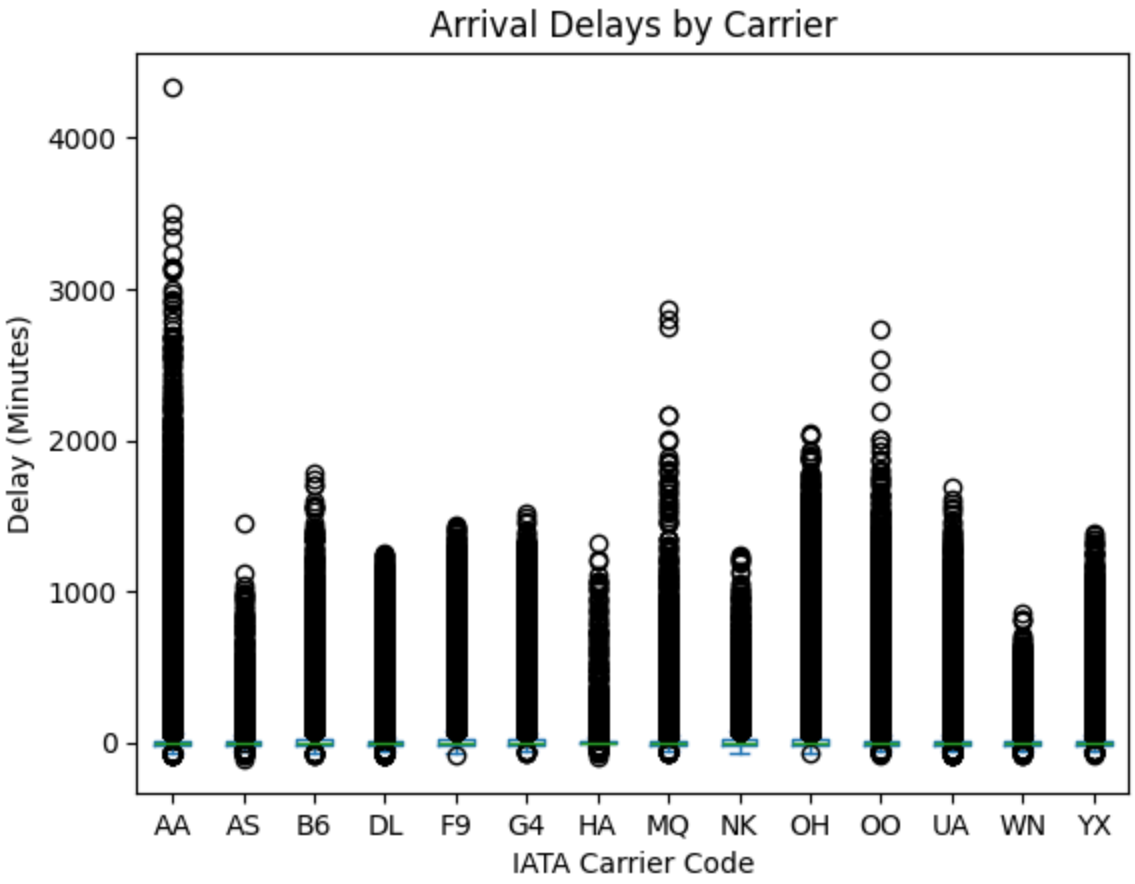
Out[15]:

	ARR_DELAY	DIVERTED
count	6.895510e+06	6895510.0
mean	8.111811e+00	0.0
std	6.065110e+01	0.0
min	-1.090000e+02	0.0
25%	-1.600000e+01	0.0
50%	-6.000000e+00	0.0
75%	1.100000e+01	0.0
max	4.336000e+03	0.0

Some of our delays are negative. According to the dataset, this means that they arrived early! Let's plot this data to draw some more conclusions.

In [16]:
```
boxplot = flights_df.plot.box(column="ARR_DELAY", by="OP_CARRIER").iloc[0]
boxplot.set_xlabel("IATA Carrier Code")
boxplot.set_ylabel("Delay (Minutes)")
boxplot.set_title("Arrival Delays by Carrier")
```

Out[16]:
```
Text(0.5, 1.0, 'Arrival Delays by Carrier')
```



It looks like this dataset has a lot of outliers, creating a lot of circles on our box plot! One flight was also delayed for about 3 days, something really bad must have happened. This is not very useful for drawing many conclusions since there are many large values. Let's use a logarithmic scale to take a closer look. Since we do have negative and zero values, we will use symlog.

In [17]:
```
carrier_colors = {
    "AA": "#0078D2",
    "AS": "#01426A",
    "B6": "#003876",
    "DL": "#C8102E",
    "F9": "#78BE20",
    "G4": "#F15A24",
    "HA": "#5F259F",
    "MQ": "#6699CC",
    "NK": "#FFD100",
    "OH": "#8C1D40",
    "OO": "#6366F1",
    "UA": "#005DAA",
    "WN": "#F9B612",
    "YX": "#607D8B"
}

boxplot = flights_df.plot.box(column="ARR_DELAY", by="OP_CARRIER", return_type="dic
plt.xlabel("IATA Carrier Code")
plt.ylabel("Delay (Minutes)")
```

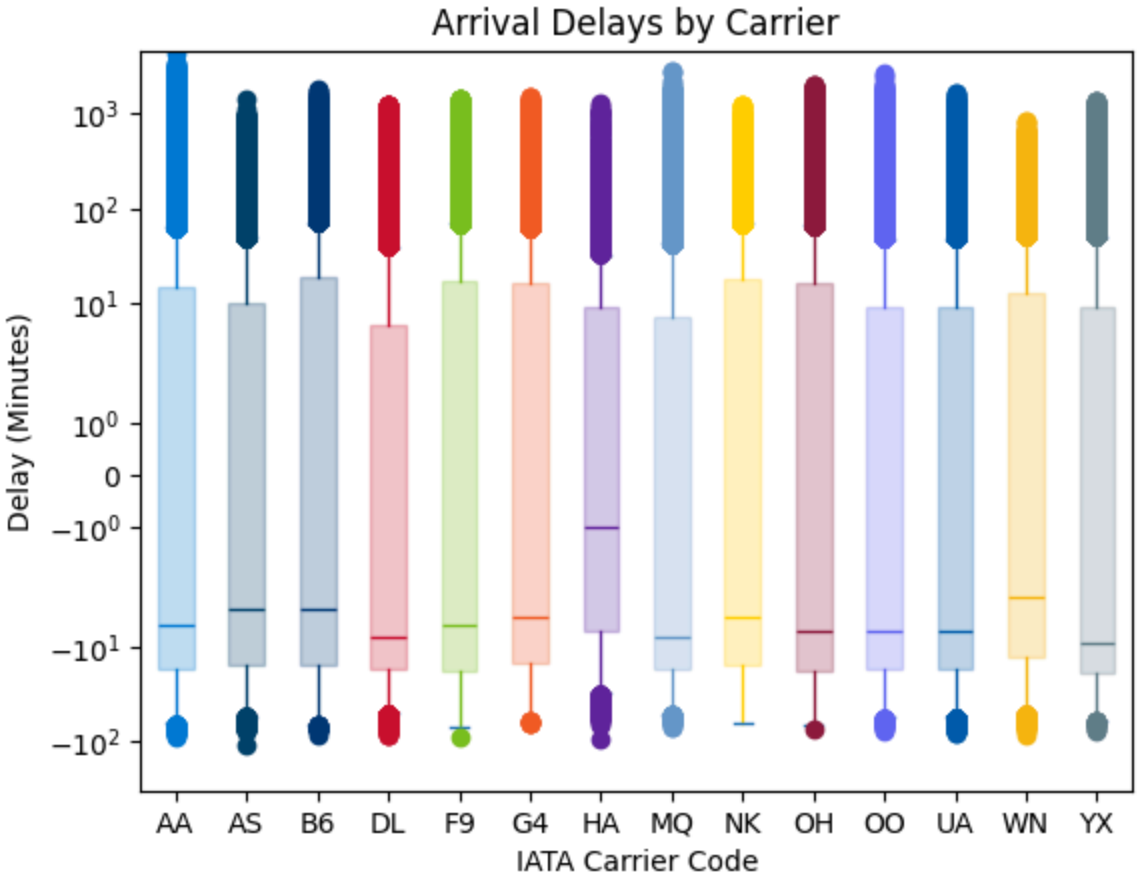
```
plt.title("Arrival Delays by Carrier")
plt.yscale('symlog')

# Make it pretty!
for i, color in enumerate(carrier_colors.values()):
    box = boxplot['boxes'][i]
    median = boxplot['medians'][i]
    fliers = boxplot['fliers'][i]

    box.set_color(color)
    box.set_alpha(0.25)
    median.set_color(color)
    fliers.set_markerfacecolor(color)
    fliers.set_markeredgecolor(color)

    # Set every 2 whiskers color
    for whisker in boxplot['whiskers'][2*i : 2*i + 2]:
        whisker.set_color(color)

plt.show()
```



The median arrival delay will be the most important here since we do have a lot of outlier data.

Conclusion 1: The median delay is negative, indicating that most flights arrive early across carriers. 75% of flights have a delay of around 10 minutes or less across carriers as well. All

carriers are planning their flights well, so you should not expect any significant delays on your flights!

American Airlines (AA) does have more outlier delays extending beyond 1,000 minutes than the other carriers. There are enough outliers across carriers that it is unlikely these datapoints were incorrectly entered, there may have been some event preventing them from taking off for more than a day.

Confidence Intervals

This subsection explores Confidence Intervals to find some statistical measure for expected delays over a time interval. This statistical measure is useful for comparing a range of mean values instead of singular hypothesees.

However, with such a large dataset, you require preprocessing in order to come up with some realistic conclusion.

In [18]:
```
# 99% confidence intervals for arrival delay (minutes) and P(delay >= 15 min) by sc

query = "SELECT DATE, ARR_DELAY, ARR_DEL15 FROM Flights WHERE DIVERTED = 0"
timestamp_delay = pd.read_sql_query(query, conn, parse_dates=['DATE'])
timestamp_delay['hour'] = timestamp_delay['DATE'].dt.hour

hourly = timestamp_delay.groupby('hour').agg(count=('ARR_DELAY', 'count'),
                                             mean=('ARR_DELAY', 'mean'),
                                             median=('ARR_DELAY', 'median'),
                                             q1=('ARR_DELAY', lambda x: x.quantile(
                                             q3=('ARR_DELAY', lambda x: x.quantile(
                                             std=('ARR_DELAY', 'std'),
                                             pct_late=('ARR_DEL15', 'mean'))  # mea
hourly['iqr'] = hourly['q3'] - hourly['q1']

# 99% CI for mean delay
se = hourly['std'] / np.sqrt(hourly['count'])
hourly['99%_lower_bound'] = hourly['mean'] - 2.576 * se
hourly['99%_upper_bound'] = hourly['mean'] + 2.576 * se

# 99% CI for P(delay >= 15 min)
p_se = np.sqrt(hourly['pct_late'] * (1 - hourly['pct_late']) / hourly['count'])
hourly['pct_late_lower'] = hourly['pct_late'] - 2.576 * p_se
hourly['pct_late_upper'] = hourly['pct_late'] + 2.576 * p_se
#pct_late_lower and pct_late_upper mean its 15+ minutes late between pct_late_lower
#with 99% confidence.
display(hourly)
```

hour	count	mean	median	q1	q3	std	pct_late	iqr	99%_lower_bound
0	10160	1.342913	-11.0	-21.0	5.00	63.202920	0.168209	26.00	-0.272323
1	3344	3.582835	-11.0	-22.0	6.00	75.568875	0.185407	28.00	0.216513
2	1251	1.358913	-12.0	-23.0	8.00	62.131370	0.204636	31.00	-3.166187
3	654	4.663609	-9.0	-19.0	6.75	70.003063	0.198777	25.75	-2.387770
4	165	3.957576	-11.0	-19.0	5.00	55.169843	0.175758	24.00	-7.106258
5	188230	-3.504038	-12.0	-20.0	-3.00	58.517411	0.083127	17.00	-3.851483
6	473284	-2.445599	-11.0	-19.0	-1.00	59.306299	0.098495	18.00	-2.667667
7	512712	-0.870949	-10.0	-18.0	1.00	54.208705	0.125376	19.00	-1.065969
8	478738	0.299968	-9.0	-18.0	3.00	50.281044	0.144323	21.00	0.112770
9	392034	1.882196	-8.0	-17.0	5.00	49.250195	0.162386	22.00	1.679572
10	439060	3.488544	-7.0	-16.0	6.00	49.581950	0.178515	22.00	3.295788
11	425918	5.250520	-7.0	-16.0	8.00	54.059134	0.197353	24.00	5.037141
12	414998	7.357811	-6.0	-16.0	10.00	58.604440	0.217430	26.00	7.123468
13	413691	8.835752	-5.0	-15.0	12.00	59.615306	0.232096	27.00	8.596990
14	391257	11.733779	-4.0	-14.0	16.00	61.823577	0.260918	30.00	11.479172
15	398252	13.040155	-3.0	-14.0	18.00	61.544199	0.276476	32.00	12.788936
16	396394	14.984387	-2.0	-14.0	21.00	63.590836	0.297674	35.00	14.724205
17	436115	16.213020	-2.0	-14.0	22.00	67.530996	0.306206	36.00	15.949600
18	404818	17.101539	-1.0	-14.0	24.00	69.756179	0.318894	38.00	16.819117
19	372684	16.964850	-1.0	-14.0	25.00	68.343683	0.322386	39.00	16.676464
20	300806	16.684863	-1.0	-14.0	25.00	66.792716	0.322640	39.00	16.371151
21	231552	14.989104	-3.0	-15.0	21.00	71.253365	0.292574	36.00	14.607663
22	151426	13.328708	-3.0	-15.0	19.00	68.680594	0.280890	34.00	12.874056
23	57967	4.506737	-7.0	-18.0	8.00	61.975084	0.197785	26.00	3.843646

Now that we have our dataframe: hourly, we can finally start making some conclusions.

We previously found reasonable suspicion that there is a difference in the amount of delays per hour and the length of the delays.

Here we must embolden our claims with Hypothesis Testing. First, let's see if departure time and the proportion of late flights are related using a Chi-Squared test, since we are working

with categorical data (late or not late). Late is defined by this dataset as a delay greater than 15 minutes.

- **Null Hypothesis:** There is no relationship between departure time and the proportion of late flights.
- **Alternative Hypothesis:** There is a relationship between departure time and the proportion of late flights.

In [19]:
```
c1 = pd.crosstab(timestamp_delay['hour'], timestamp_delay['ARR_DEL15'])
chi2, p, df, _ = stats.chi2_contingency(c1)
print(f"Chi-squared = {chi2:.1f}, dof = {df}, p = {p}")
display(c1)
```

```
Chi-squared = 233363.2, dof = 23, p = 0.0
```

ARR_DEL15	0.0	1.0
hour		
0	8451	1709
1	2724	620
2	995	256
3	524	130
4	136	29
5	172583	15647
6	426668	46616
7	448430	64282
8	409645	69093
9	328373	63661
10	360681	78379
11	341862	84056
12	324765	90233
13	317675	96016
14	289171	102086
15	288145	110107
16	278398	117996
17	302574	133541
18	275724	129094
19	252536	120148
20	203754	97052
21	163806	67746
22	108892	42534
23	46502	11465

Conclusion 2A: The p value is 0.0 (< .05), which shows there's a statistical significance and we reject the null hypothesis that there is no relationship between departure time and the proportion of late flights. However, because we are working with a large dataset and contingency table, Chi-squared is likely to show a statistical significance even for a very small difference. Let's apply Cramér's V, which will give us a value 0-1 showing how strong the association is.

In [20]:
```
stats.contingency.association(c1, method="cramer")
```

Out[20]:
```
0.18396405037696714
```

Conclusion 2B: Cramér's V gave a value of 0.184, showing there is a slight association between departure time and proportion of late flights.

Let's narrow our focus and examine the difference in proportion of late flights in the morning (6 AM - 11 AM) compared to the evening (5 PM - 10 PM). To do this, we will use a pooled proportion z-test. This test will work since our sample size is large enough (millions of flights).

- **Null Hypothesis:** The proportion of late flights is the same in the morning and evening.
- **Alternative Hypothesis:** Evening flights have a higher proportion of late flights than morning flights (p1 > p2).

In [21]:
```
# morning (6-11) vs evening (17-22)... is the evening rate higher?
morning = timestamp_delay[timestamp_delay['hour'].between(6, 11)]['ARR_DEL15'].drop
evening = timestamp_delay[timestamp_delay['hour'].between(17, 22)]['ARR_DEL15'].dro

p_morning = morning.mean()
p_evening = evening.mean()
n_morning = morning.count()
n_evening = evening.count()

# pooled-proportion assuming that the null hypo is true... that rates are equal.
p_pool = (morning.sum() + evening.sum()) / (n_morning + n_evening)
se = np.sqrt(p_pool * (1 - p_pool) * (1/n_morning + 1/n_evening))

z = (p_evening - p_morning) / se # p1 = p_evening, p2 = p_morning
p = stats.norm.sf(z)  # Right-tailed test (p1 > p2)

print(f"Morning late rate: {p_morning:.1%}, Evening late rate: {p_evening:.1%}")
print(f"z = {z:.2f}, p = {p}")
```

```
Morning late rate: 14.9%, Evening late rate: 31.1%
z = 416.00, p = 0.0
```

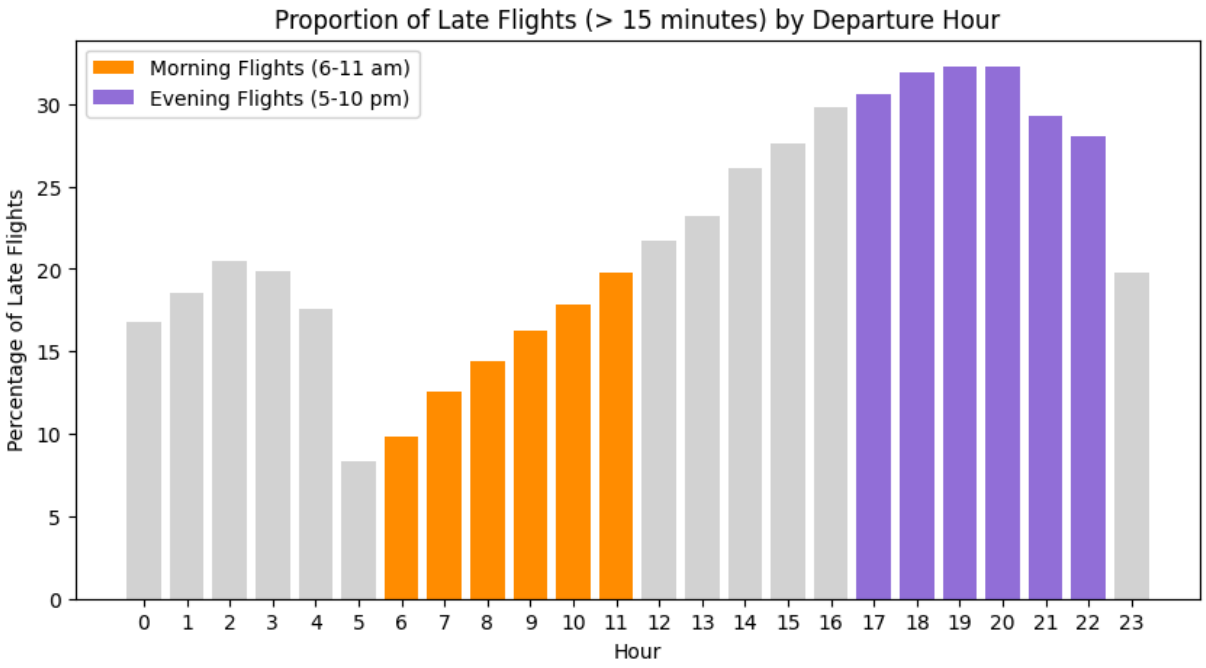
Conclusion 3A: Once again, we have a p value of 0 (< .05), rejecting the null hypothesis that the proportion of late flights is the same in the morning and evening. Looking at the rate, this information is not surprising since the rate of late flights in the evening is 31%, compared to 15% in the morning. To confirm our suspicions, we can look at a graph:

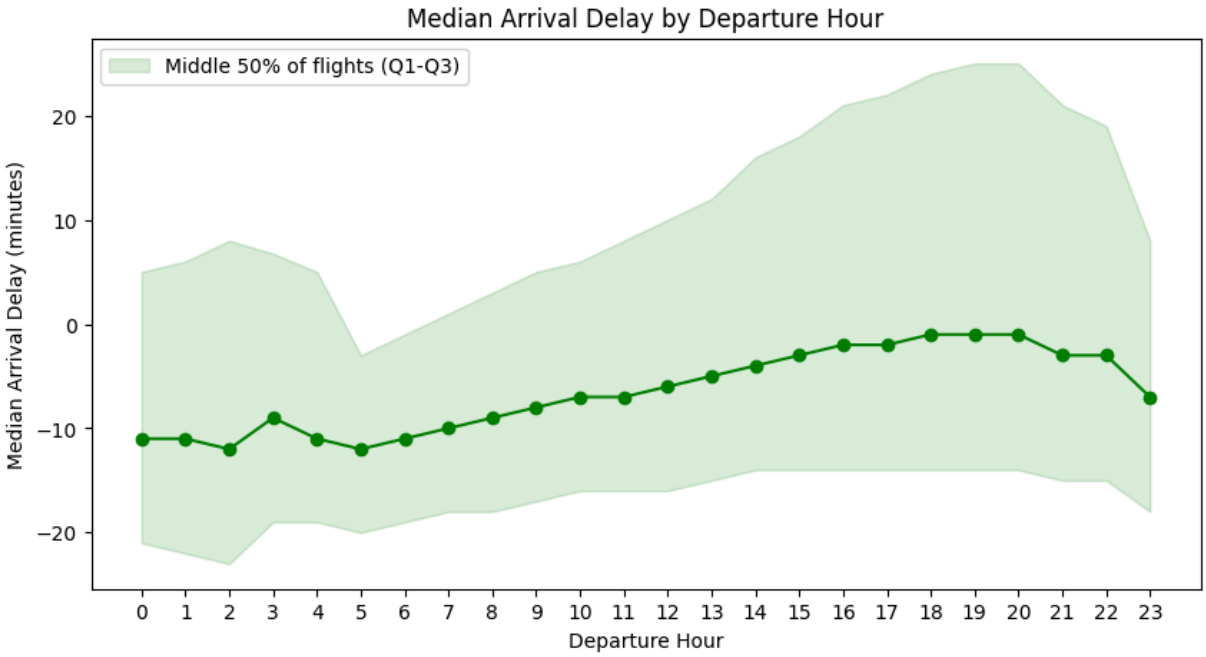
In [22]:
```
# Morning vs Evening Delays
segment_colors = {
    "morning": "darkorange",
    "evening": "mediumpurple",
    "other": "lightgray"
}
```

```
colors = [
    segment_colors["morning"] if 6 <= h <= 11
    else segment_colors["evening"] if 17 <= h <= 22
    else segment_colors["other"]
    for h in hourly.index
]

plt.figure(figsize=(10, 5))
plt.bar(hourly.index, hourly['pct_late'] * 100, color=colors)
plt.xticks(range(24))
plt.xlabel('Hour')
plt.ylabel('Percentage of Late Flights')
plt.title('Proportion of Late Flights (> 15 minutes) by Departure Hour')
legend_items = [
    Patch(facecolor=segment_colors["morning"], label="Morning Flights (6-11 am)"),
    Patch(facecolor=segment_colors["evening"], label="Evening Flights (5-10 pm)"),
]
plt.legend(handles=legend_items)
plt.show()

# Medians plot with IQR band
plt.figure(figsize=(10,5))
plt.plot(hourly.index, hourly['median'], color='green', marker='o')
plt.fill_between(hourly.index, hourly['q1'], hourly['q3'], color='green', alpha=0.1
plt.xticks(range(24))
plt.xlabel('Departure Hour')
plt.ylabel('Median Arrival Delay (minutes)')
plt.title('Median Arrival Delay by Departure Hour')
plt.legend()
plt.show()
```





Conclusion 3B: Looking at the proportion of late flights in the first graph, we can clearly see a trend that the proportion of late flights in the evening is greater compared to the morning. This confirms our earlier analysis.

Conclusion 2C: The second graph is measuring the median and Q1 - Q3 arrival delay based on the departure hour. We can see there is a general trend of increasing delays as the day progresses from 5 AM until around 8 PM, which could be related to us seeing a statistically significant difference between the proportion of late flights and various departure times.

Exploring the Relationship Between Time of Year and Weather Delays

As weather across the county often varies based on the time of year, it can be useful to examine how the frequency of weather delays might be affected by what time of year flights are occurring. To explore this relationship, we can use hypothesis testing.

- **Null Hypothesis:** There is no significant relationship between the time of year and the frequency of weather delays.
- **Alternative Hypothesis:** There is a significant relationship between the time of year and the frequency of weather delays.

To test these hypotheses, we can leverage certain columns of the dataset. The dataset already contains a 'QUARTER' column, which represents which quarter of the year each flight occurred, with quarter 1 being January-March, quarter 2 being April-June, and so on. The 'WEATHER_DELAY' column measures the weather delay for each flight in minutes. While most flights have little to no weather delay, we can define a weather-delayed flight as a

flight that has a 'WEATHER_DELAY' of 15 minutes or more. Examining the frequency of weather delays for each quarter can allow us to test our hypotheses.

To start, we can query our SQL data table to find the frequency of weather-delayed flights and non-weather-delayed flights, grouped by quarter:

In [23]:
```
conn = sqlite3.connect("database.sqlite")

delays = pd.read_sql("SELECT quarter,COUNT(weather_delay) AS delayed FROM Flights W
non_delays = pd.read_sql("SELECT quarter,COUNT(weather_delay) AS not_delayed FROM F
```

Then, we can combine these two tables with an inner join based on 'QUARTER' to create a complete contingency table to compare quarter vs. weather delay state:

In [24]:
```
frequencies = delays.set_index('QUARTER').join(non_delays.set_index('QUARTER'), on=
display(frequencies)
```

QUARTER	delayed	not_delayed
1	17410	1598138
2	17566	1777929
3	20806	1749350
4	13845	1718073

Now that we have a contingency table, we can apply our chosen hypothesis test. As we are testing categorical data (where the categories are each quarter and each state of delay/non-delay), we will use the chi-squared hypothesis test:

In [25]:
```
result = stats.chi2_contingency(frequencies)
print(f"chi-squared test p-value: {result.pvalue}")
```

```
chi-squared test p-value: 6.7437184435524905e-292
```

As we can see, the p-value is much lower than $\alpha = 0.05$, so we can reject the Null Hypothesis and conclude that there is a significant relationship between the time of year and the frequency of weather delays.

However, to get a better understanding of what this means, we must use Post-Hoc testing to evaluate how specific quarters differ from each other in the frequency of weather delays. To do this, we can apply a pairwise chi-squared test to each possible pair of quarters, and use their p-values to evaluate the relationship for each pair. As we are doing multiple pairwise tests, we need to apply the Bonferroni correction to the p-values to prevent Type I errors.

In [26]:
```
pairs = list()
pvalues = list()
```

```
# do a pairwise chi2 test for each combination of quarters to compare
for pair in combinations(frequencies.index, 2):
    pair_result = stats.chi2_contingency(frequencies.loc[list(pair)])
    pairs.append(pair)
    pvalues.append(pair_result.pvalue)

# apply the Bonferroni correction to prevent Type I errors occurring from running m
reject, pvalues, _, _ = multipletests(pvalues, method='bonferroni')

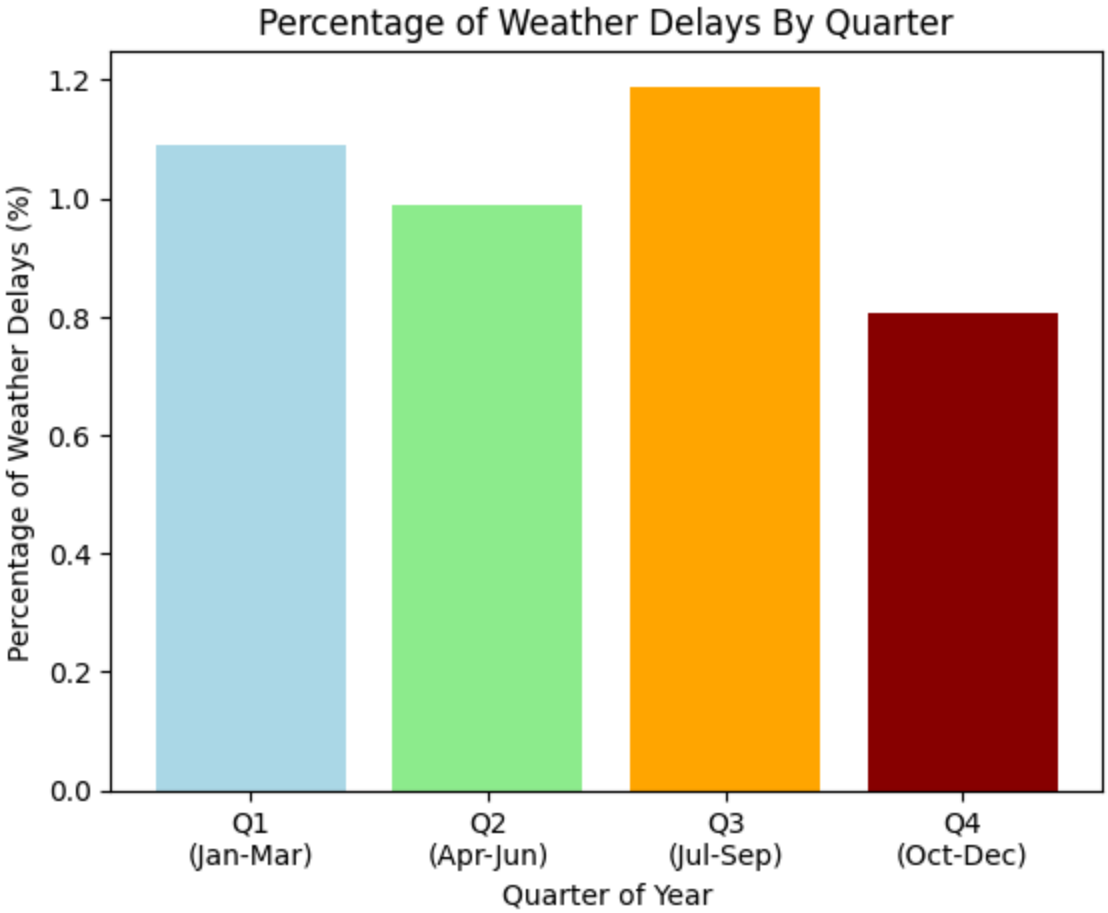
# print the adjusted p-value and verdict for each pair
for i in range(len(pairs)):
    print(f"Post-hoc test on pair {pairs[i]}: p-value = {pvalues[i]:10.4g}, reject:
```

```
Post-hoc test on pair (1, 2): p-value =  6.175e-19, reject: True
Post-hoc test on pair (1, 3): p-value =  1.171e-16, reject: True
Post-hoc test on pair (1, 4): p-value = 2.502e-153, reject: True
Post-hoc test on pair (2, 3): p-value =  7.475e-72, reject: True
Post-hoc test on pair (2, 4): p-value =  1.054e-70, reject: True
Post-hoc test on pair (3, 4): p-value = 8.348e-276, reject: True
```

Conclusion 4A: As we can see, the adjusted p-value for each pairwise combination of quarters is less than $\alpha = 0.05$, and the Null Hypothesis is rejected for each. Thus, there is a statistically significant difference in the frequency of weather delays between every pair of quarters.

To visually see this pattern, we can use a bar graph which contains the percentage of flights that are delayed in each quarter of the year:

In [27]:
```
plt.bar(["Q1\n(Jan-Mar)", "Q2\n(Apr-Jun)", "Q3\n(Jul-Sep)", "Q4\n(Oct-Dec)"], frequ
plt.title("Percentage of Weather Delays By Quarter")
plt.ylabel("Percentage of Weather Delays (%)")
plt.xlabel("Quarter of Year")
plt.show()
```



Conclusion 4B: From this graph, we can very clearly see that there is a slight difference in the percentage of weather delays between each quarter, confirming our earlier analysis that there is a statistically significant difference.